Servlet Request Dispatcher and Send Redirect

Problem Statement

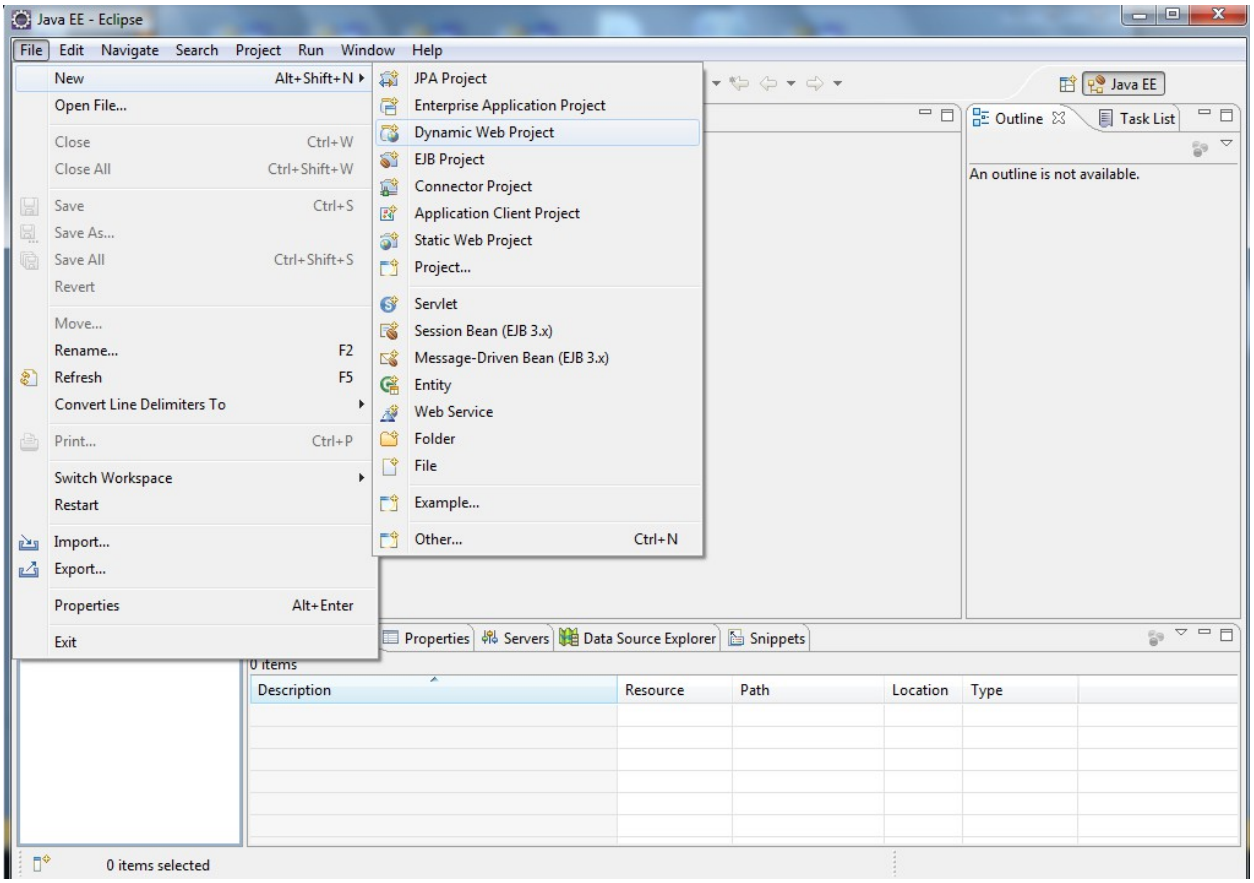
Develop a simple java web application to implement Request Dispatcher and Send Redirect.The system should get the customer name and based on the customer name should perform the following functionalities.

1) If the customer name is not tcs, a welcome message should be shown to the user
2) If the customer name is tcs, it should be redirected to Ultimatix home page
3) If the customer name is empty, the system should prompt an error message in the same page.

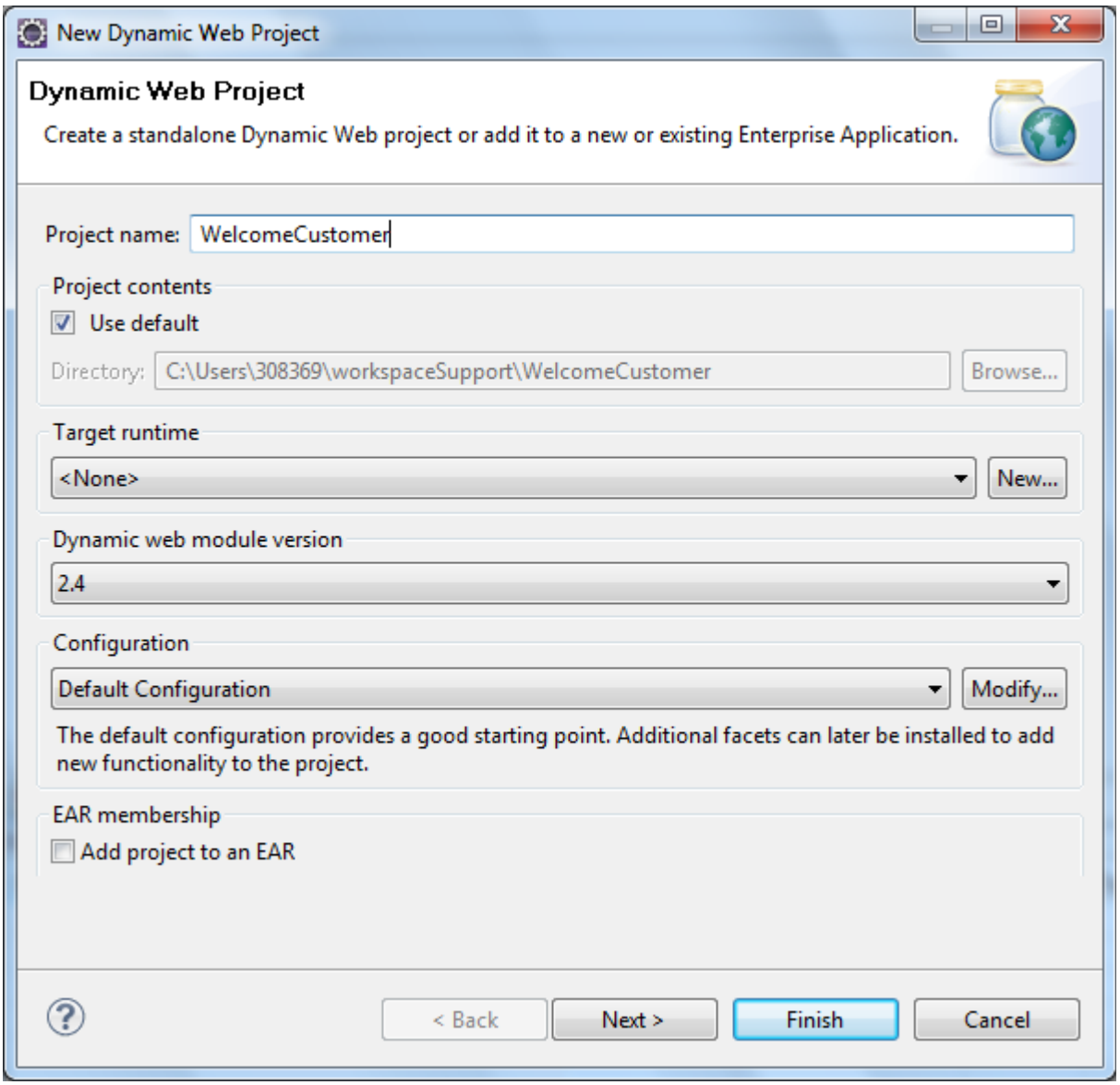
Use HTML for developing view component and servlet for controllers.

1) WelcomeCustomer

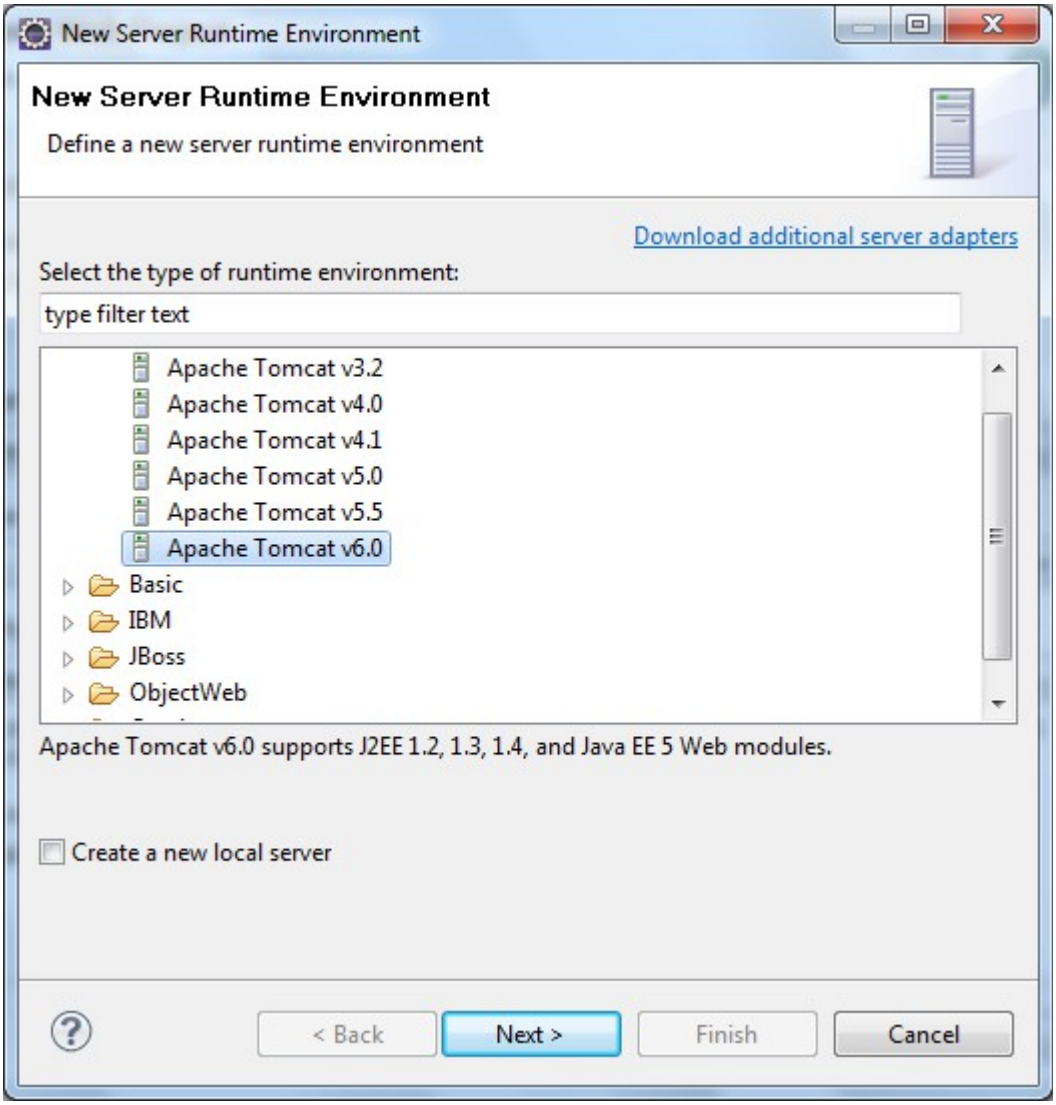
Create dynamic Web project **by File->New->Dynamic Web Project**



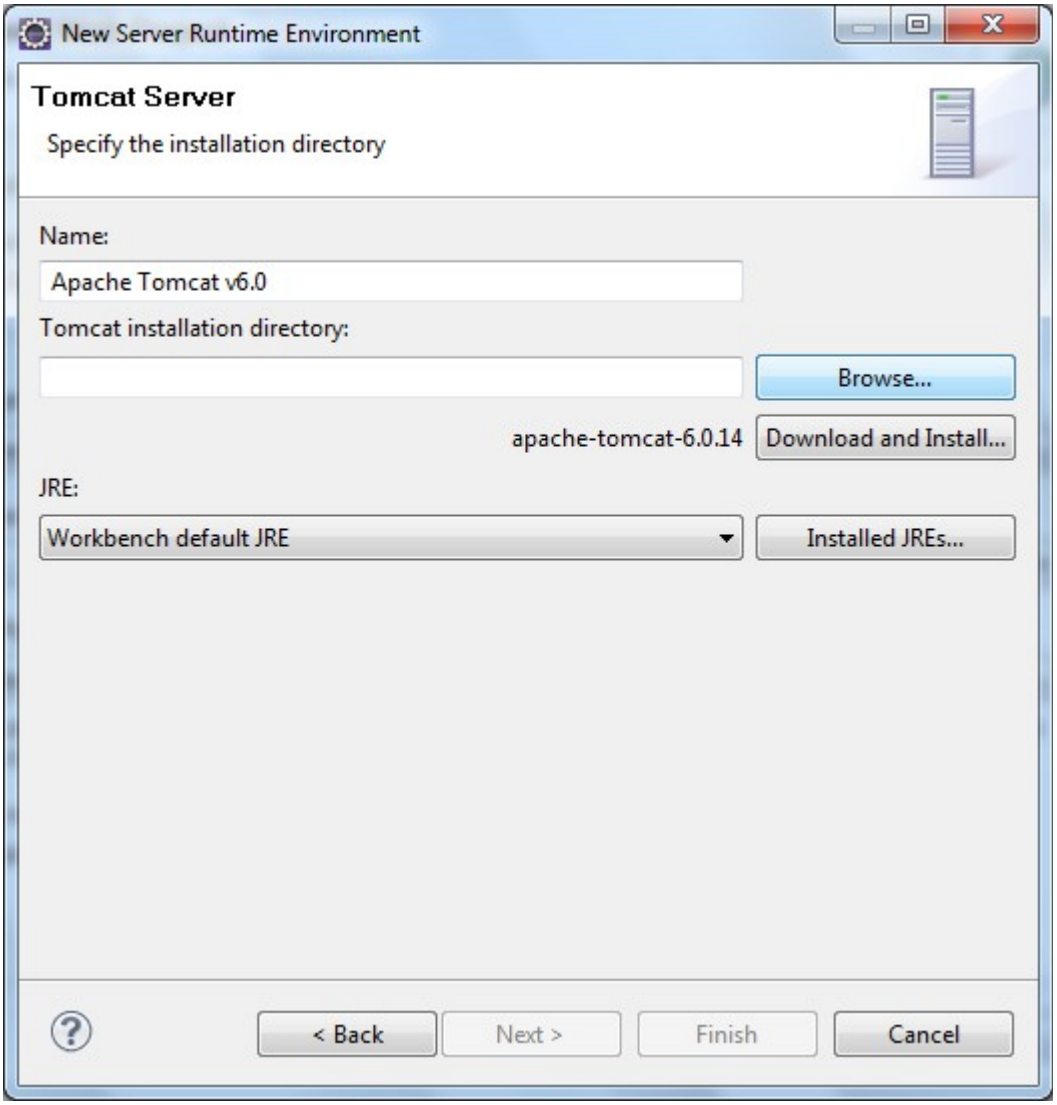
Provide a valid project name "WelcomeCustomer" adhering to the coding standards



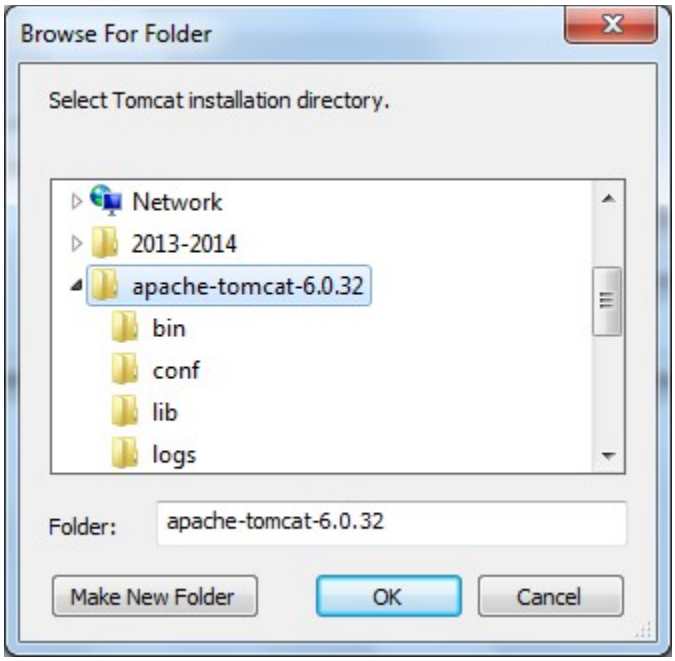
Set the new target runtime environment by clicking on New in Target runtime. Select **Apache Tomcat v6.0** and click on Next button.



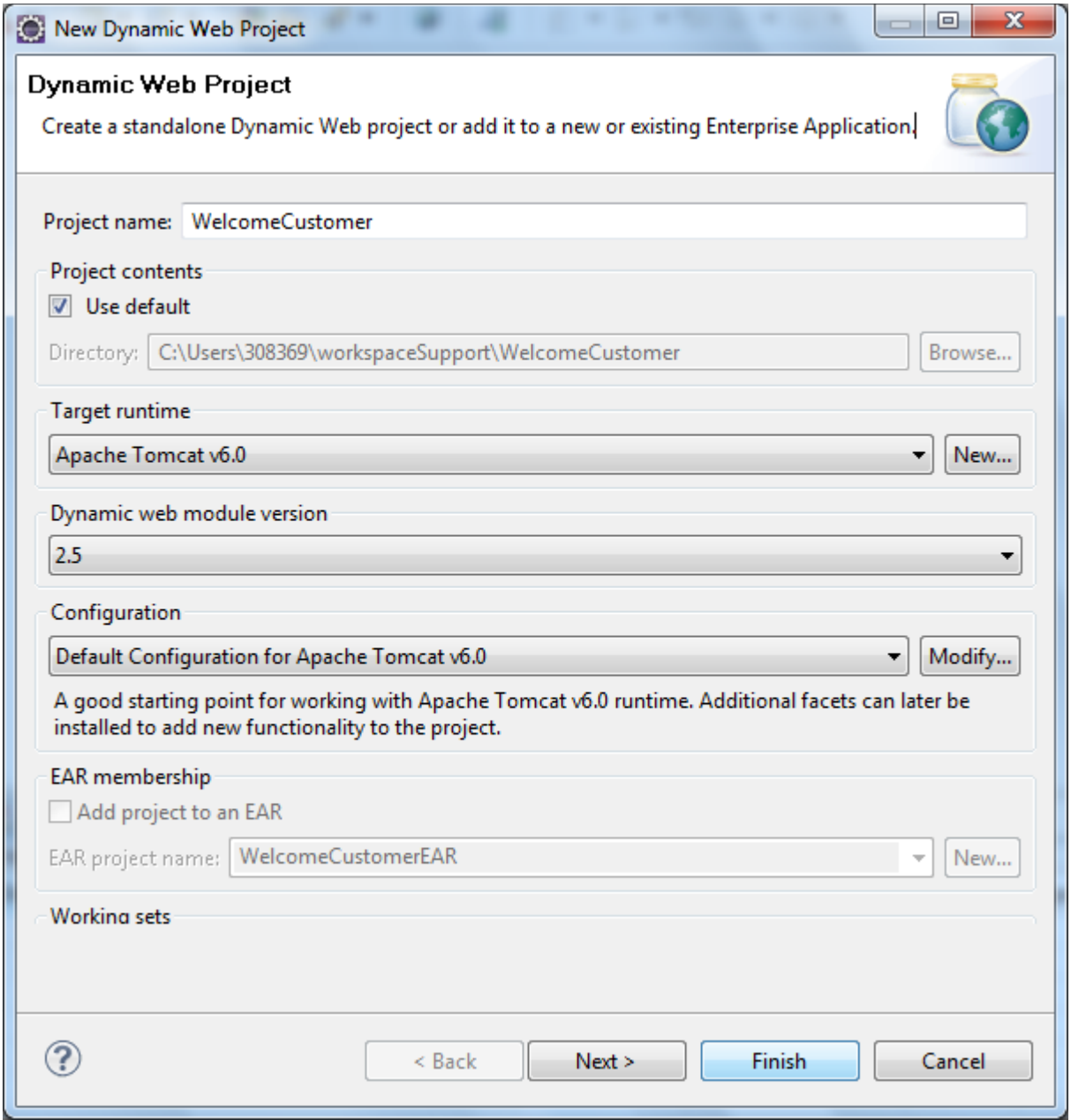
Browse for the right installation directory for Apache Tomcat v6.0



Browse for the right folder of apache-tomcat-6.032 from the local and click on OK

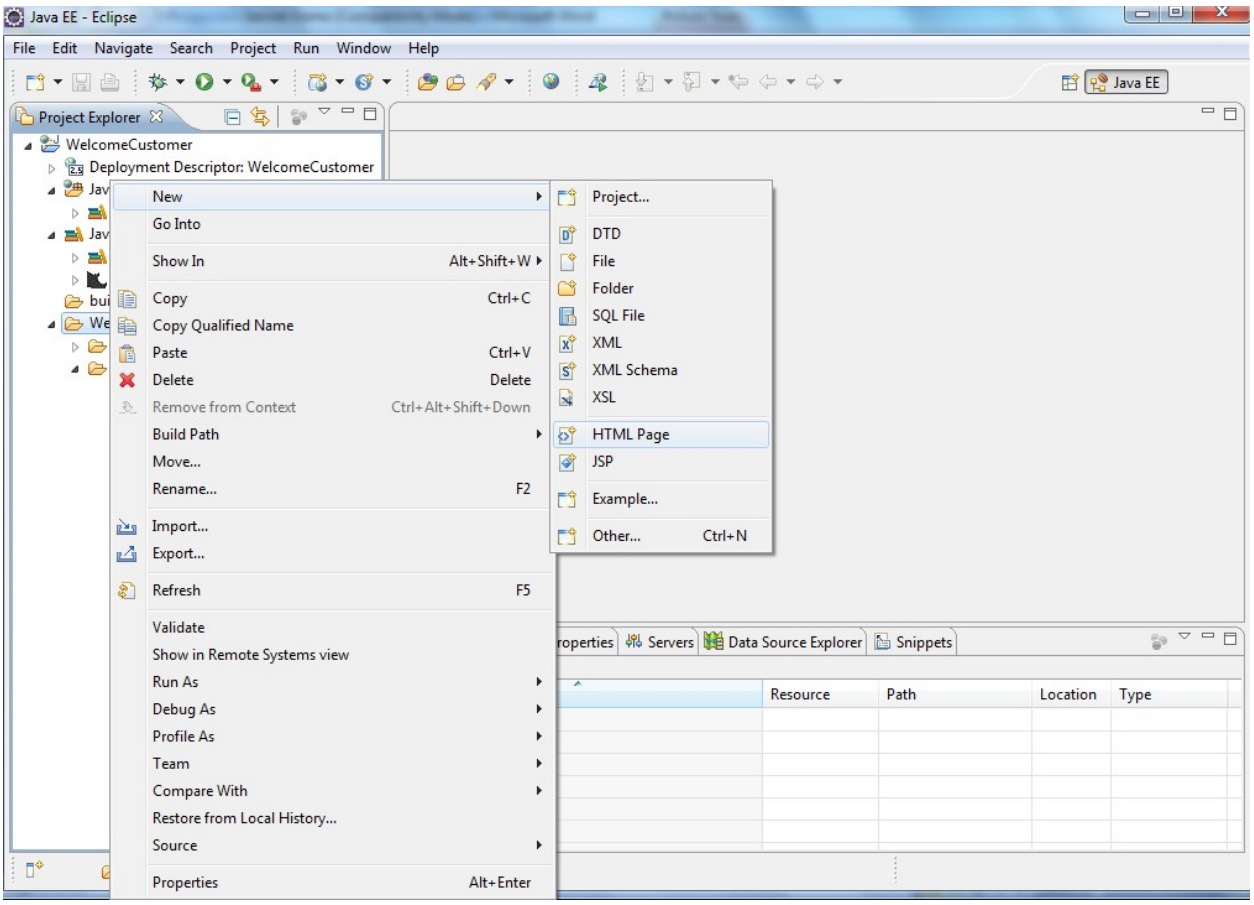


Once the target runtime gets populated click on finish.
A dynamic Web Project “WelcomeCustomer” with the folder structure as follows gets created. The same can be viewed in the project explorer.

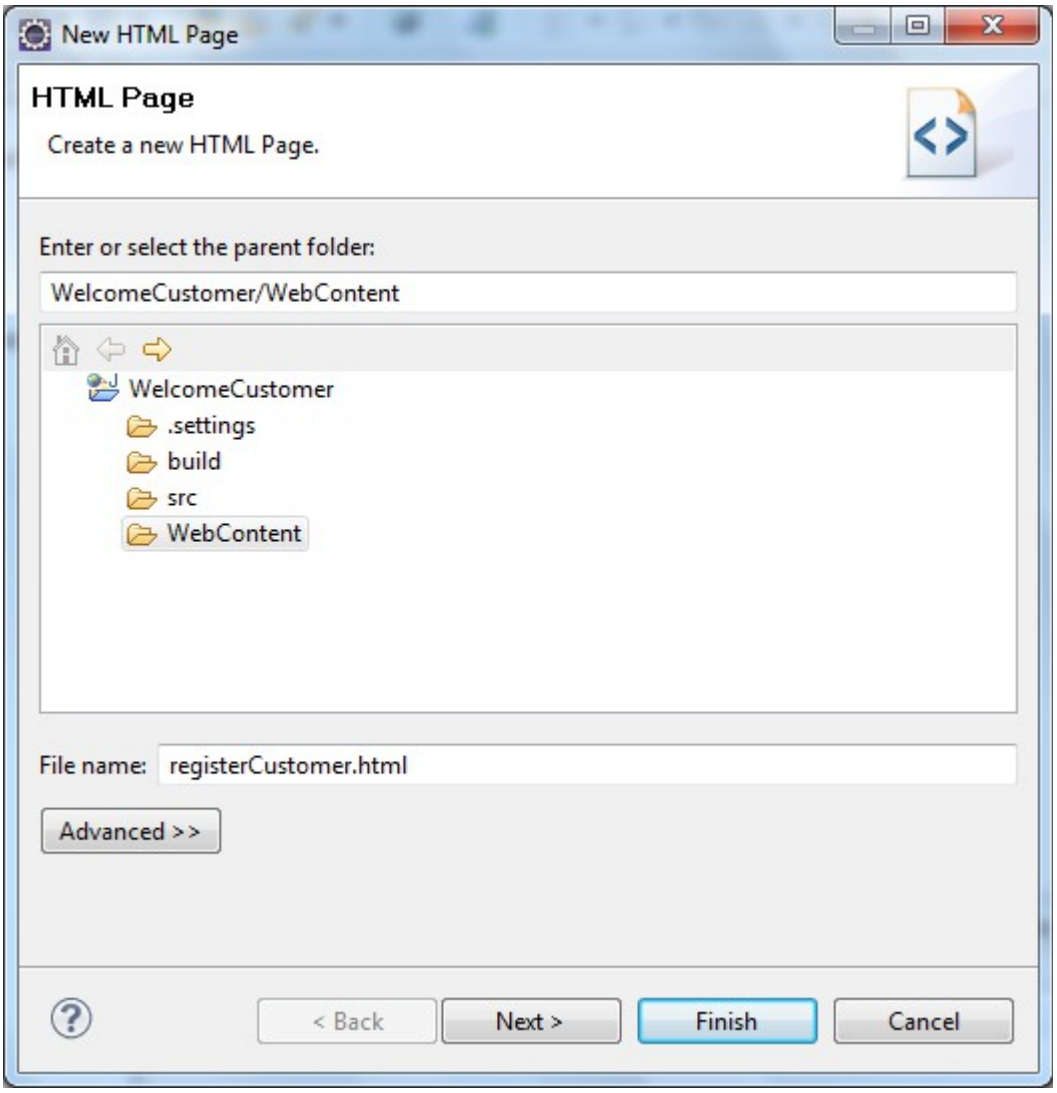


2) registerCustomer.html

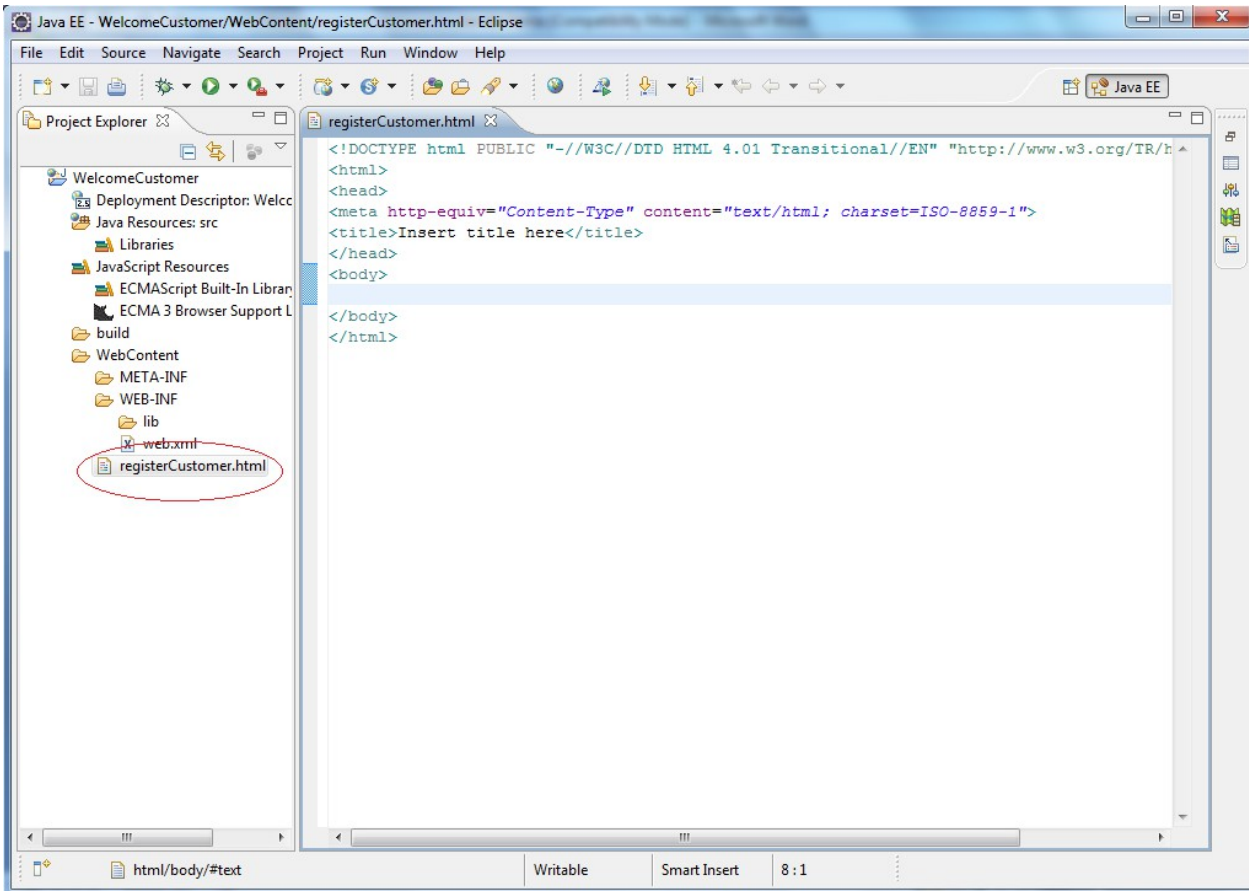
Create registerCustomer.html inside Webcontent of the Dynamic Web project WelcomeCustomer.



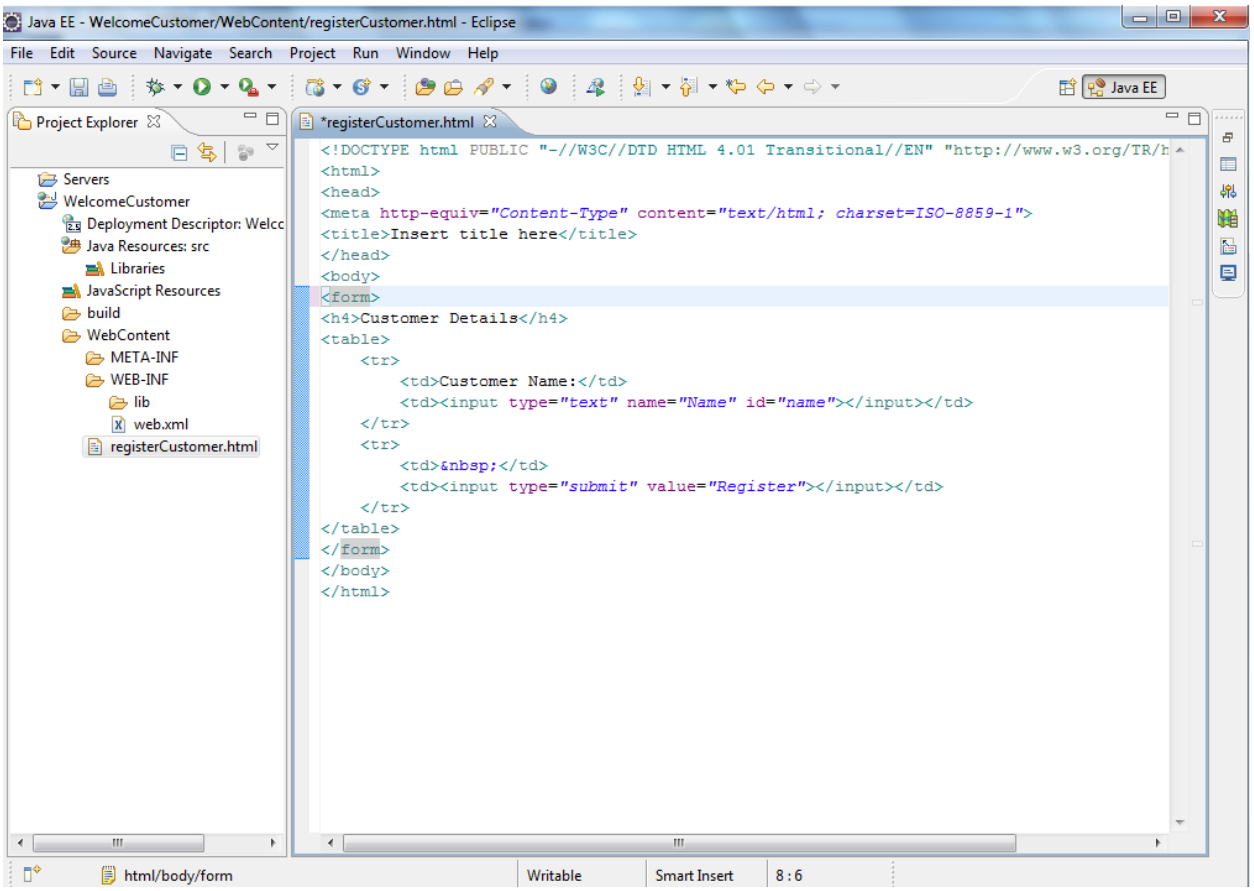
Provide a valid name for the html page and click on ‘Finish’.



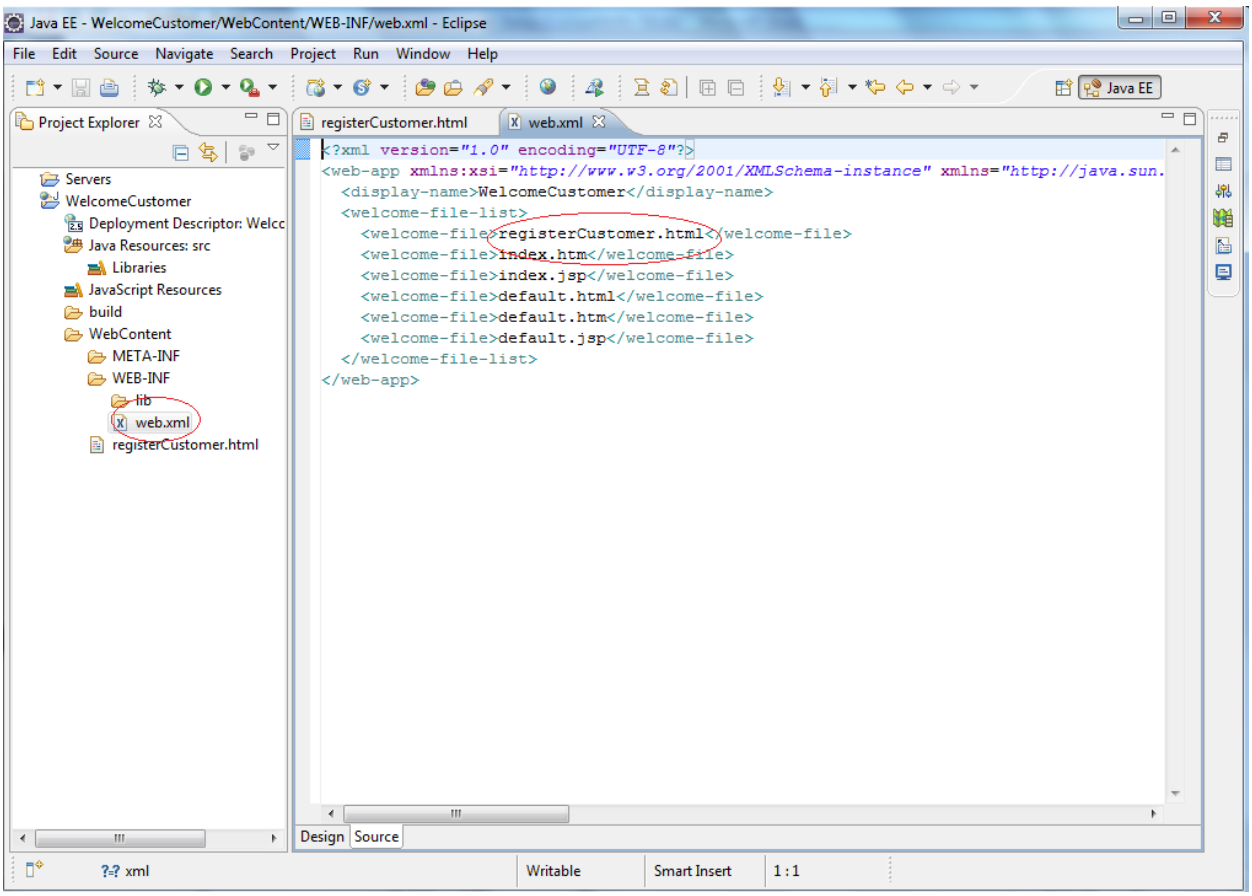
“registerCustomer.html” gets created inside the WebContent folder as highlighted



Create a form inside the registerCustomer.html that wlll get the Customer name.

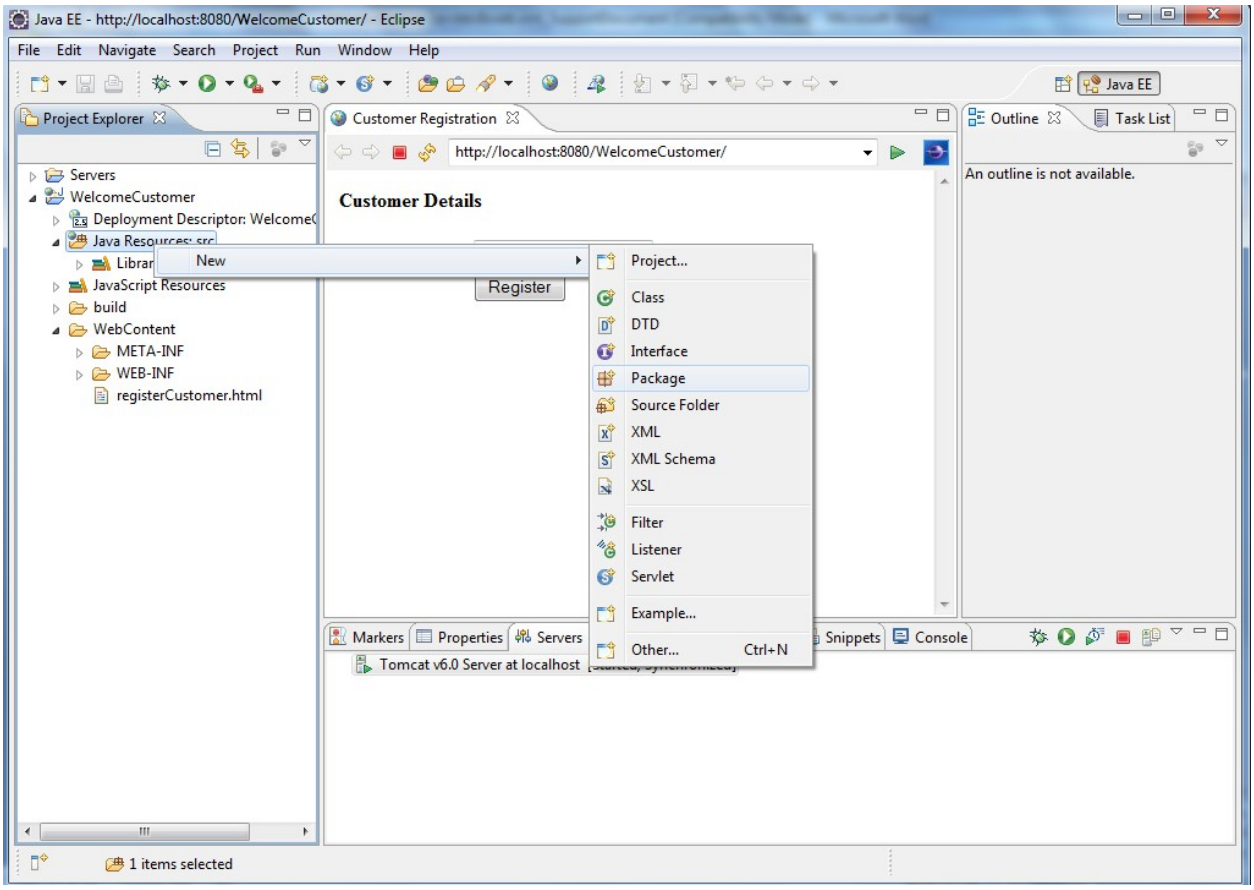


Provide the welcome file name as registerCustomer.html inside web.xml

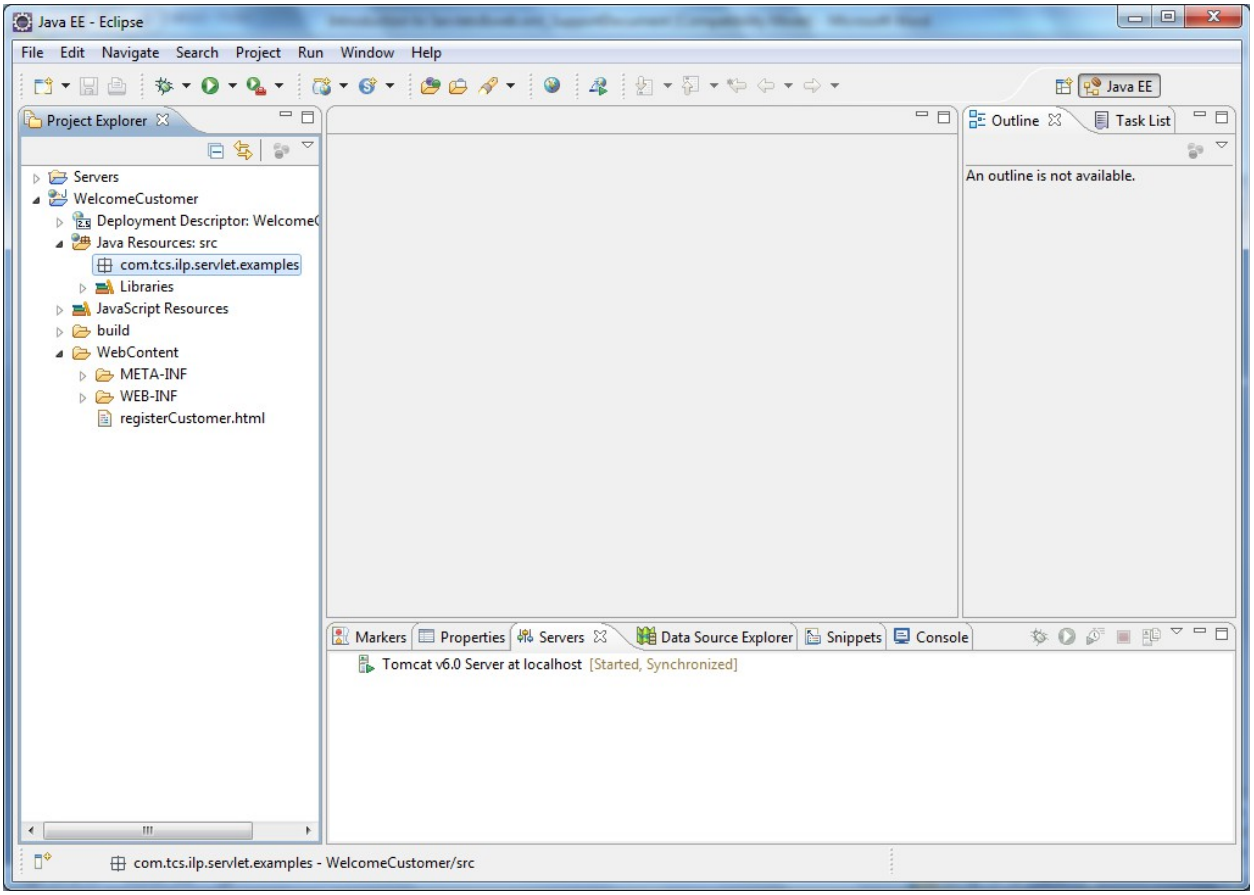


3) CustomerController.java

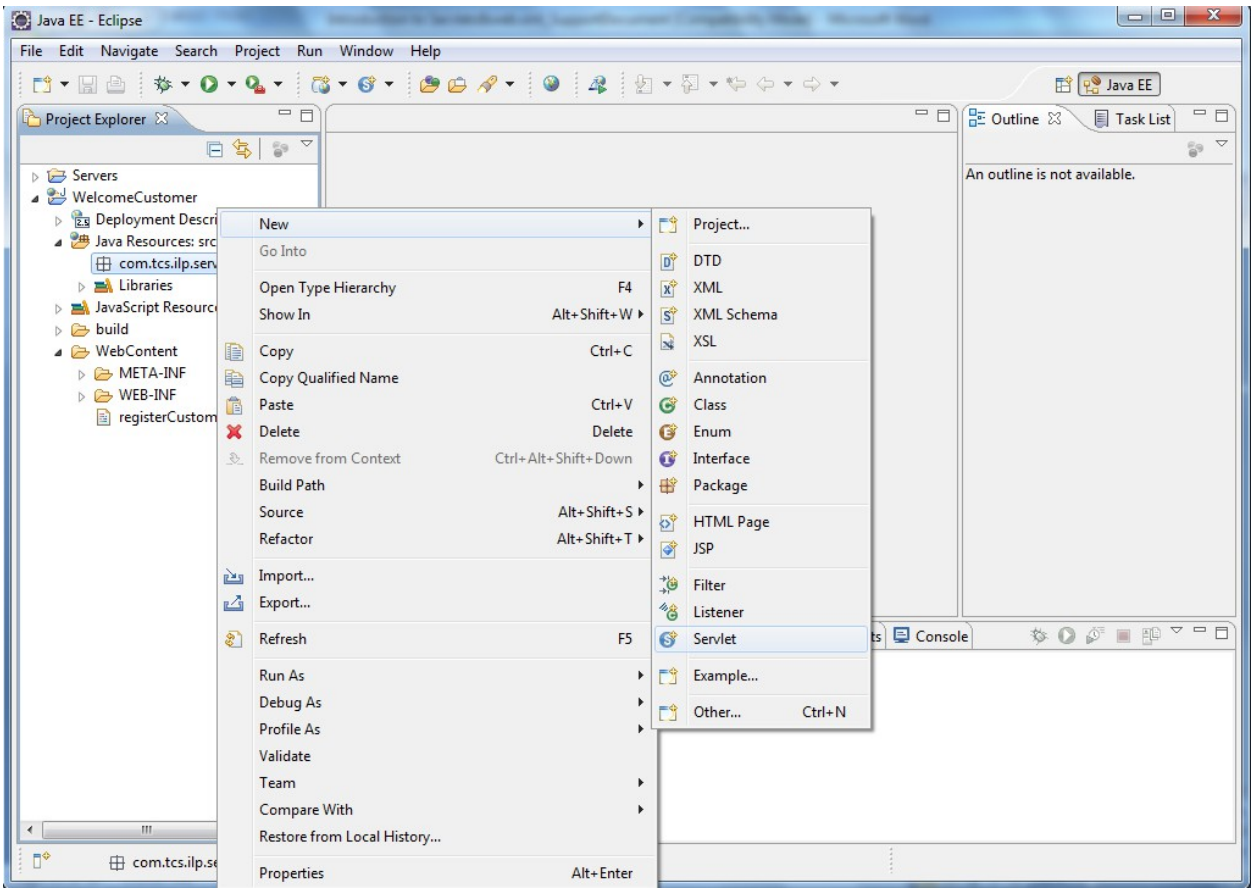
Create a package com.tcs.ilp.servlet.examples inside src folder



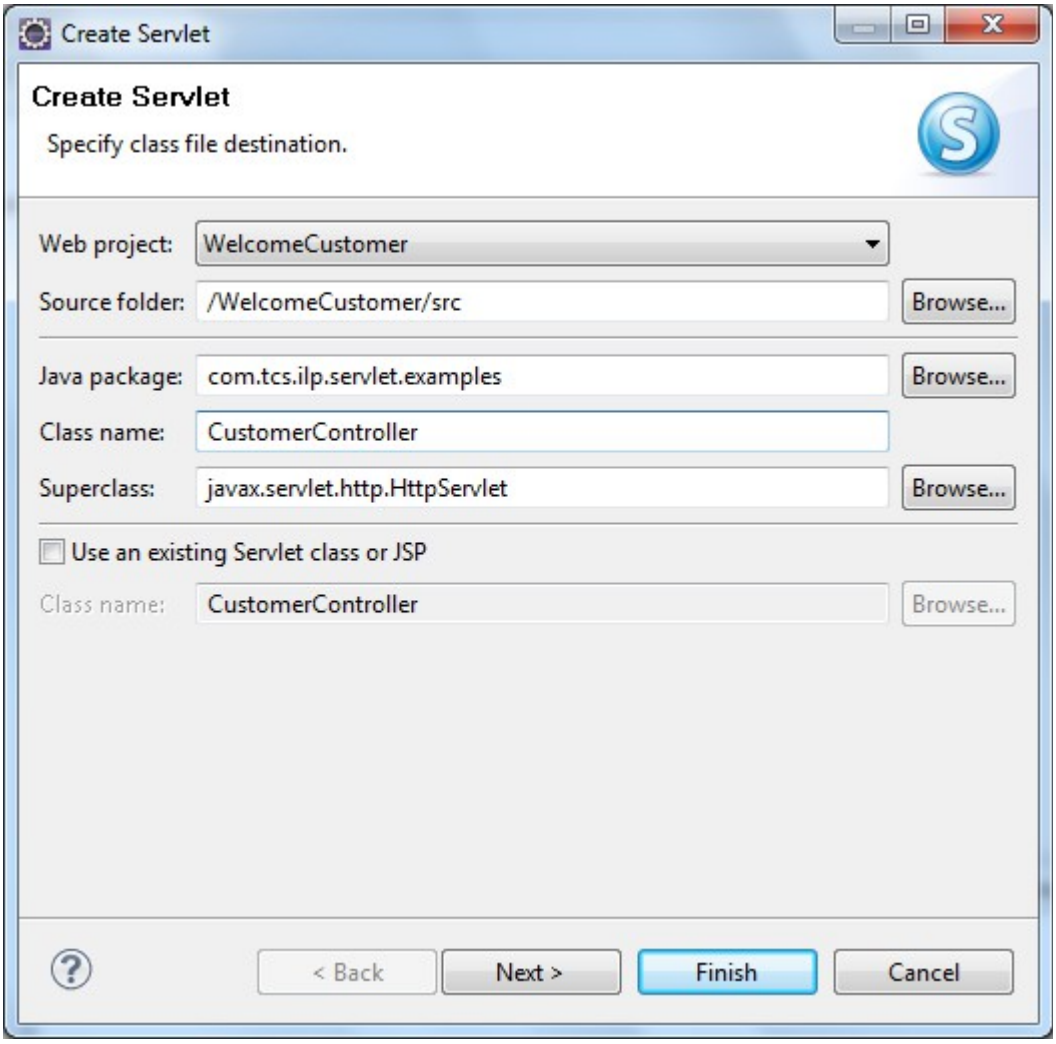
The package gets created inside the src folder



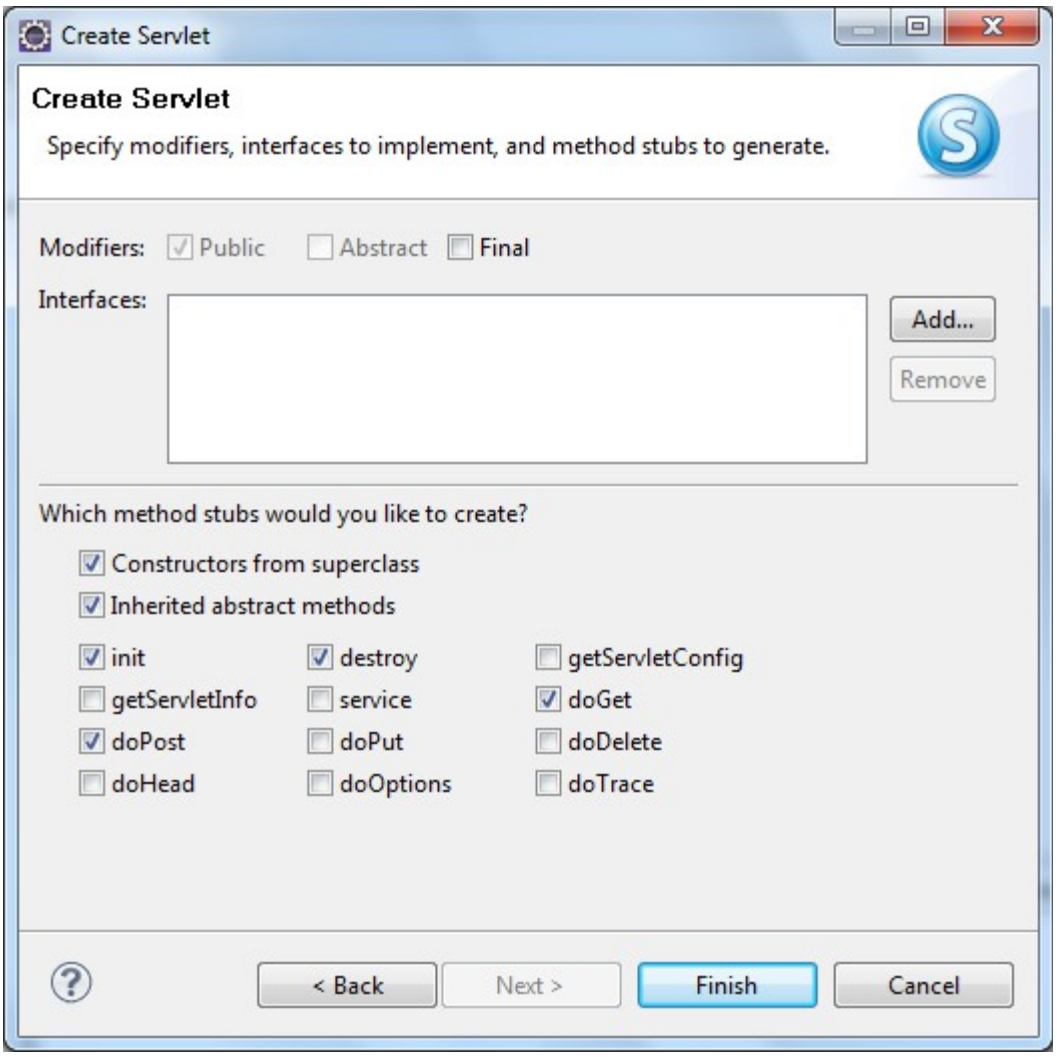
Create a new Servlet CustomerController inside the package.



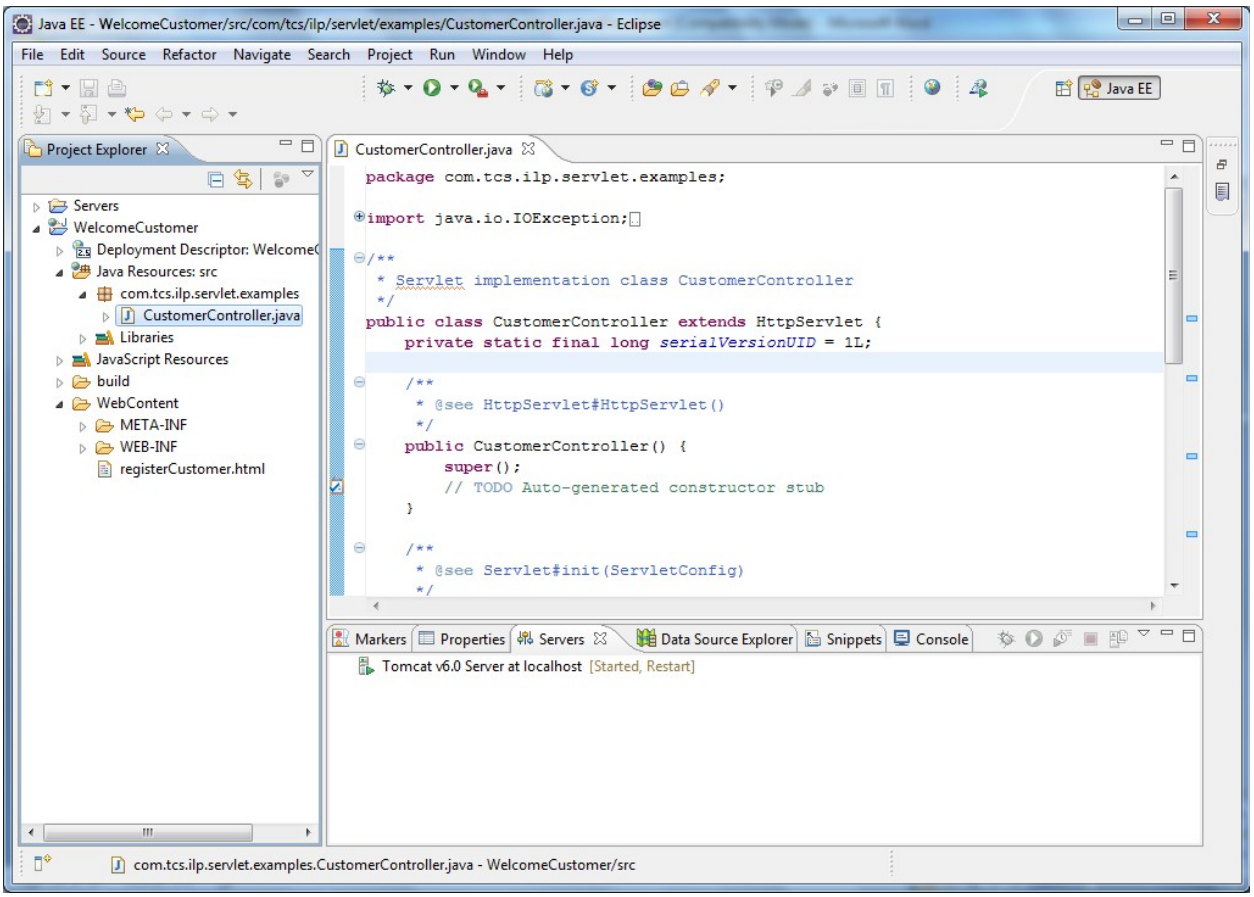
Provide a valid classname for the Servlet and click on Next



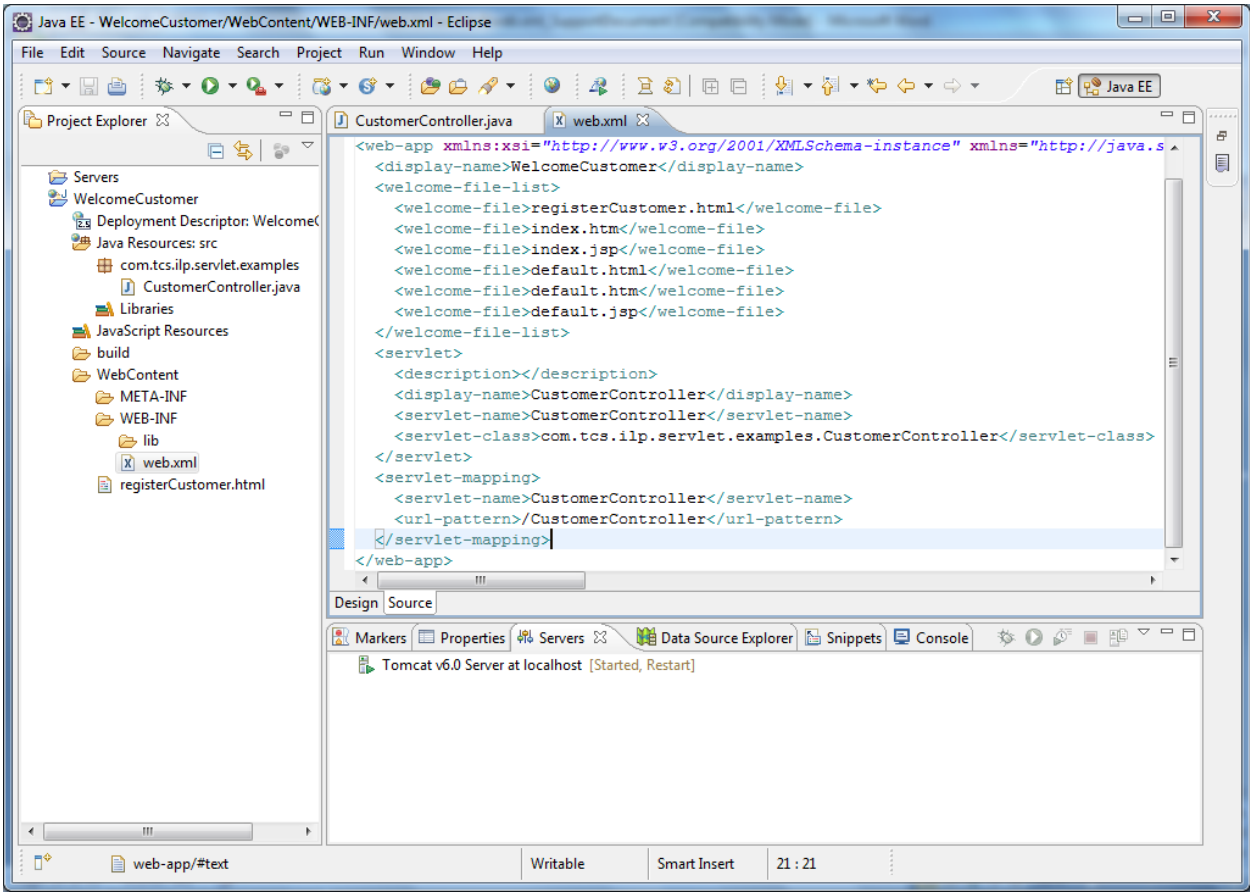
Try to override the lifecycle methods init, destroy,doGet and doPost and click on Finish



The servlet gets created under the package structure



As soon as the servlet is created the corresponding servlet mapping gets created inside the web.xml
Servlet definition has the display name, servlet name and the servlet-class it which it is available
Servlet mapping has the servlet name and the url-pattern.



Code for retrieving the customer keyed in name through request is provided inside servlet

Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/CustomerController.java - Eclipse

```java
package com.tcs.ilp.servlet.examples;

import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletConfig;
import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;
import javax.servlet.RequestDispatcher;

/**
 * Servlet implementation class CustomerController
 */
public class CustomerController extends HttpServlet {
    private static final long serialVersionUID = 1L;

    /**
     * @see HttpServlet#HttpServlet()
     */
    public CustomerController() {
        super();
        // TODO Auto-generated constructor stub
    }

    /**
     * @see Servlet#init(ServletConfig)
     */
    public void init(ServletConfig config) throws ServletException {
        // TODO Auto-generated method stub
        System.out.println("Inside Servlet initilisation");
```

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The customer name is got from the request.

- If the Customer name is not TCS and not empty, the page will be redirected to new servlet WelcomeServlet which is going to hold the welcome message.
- If the Customer name is TCS, the page is forwarded to Ultimatix home page
- If the customer name is empty, the user will get a error message prompting to enter the customer name.

Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/CustomerController.java - Eclipse

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CustomerController.java

```
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        doPost(request,response);
    }

    /**
     * @see HttpServlet#doPost(HttpServletRequest request, HttpServletResponse
     *      response)
     */
    protected void doPost(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        String customerName= request.getParameter("Name");
        PrintWriter out = response.getWriter();


        if (!(customerName.equalsIgnoreCase("TCS")) && customerName!="") {
            RequestDispatcher rd = request
                    .getRequestDispatcher("WelcomeServlet");
            rd.forward(request, response);
        } else if (customerName.equalsIgnoreCase("TCS")) {
            response.sendRedirect("https://www.ultimatix.net/");
        } else {
            RequestDispatcher rd = request.getRequestDispatcher("registerCustomer.html");
            out.print("<b>Sorry! Please enter a customer name</b>");
            rd.include(request, response);
        }

    }
```

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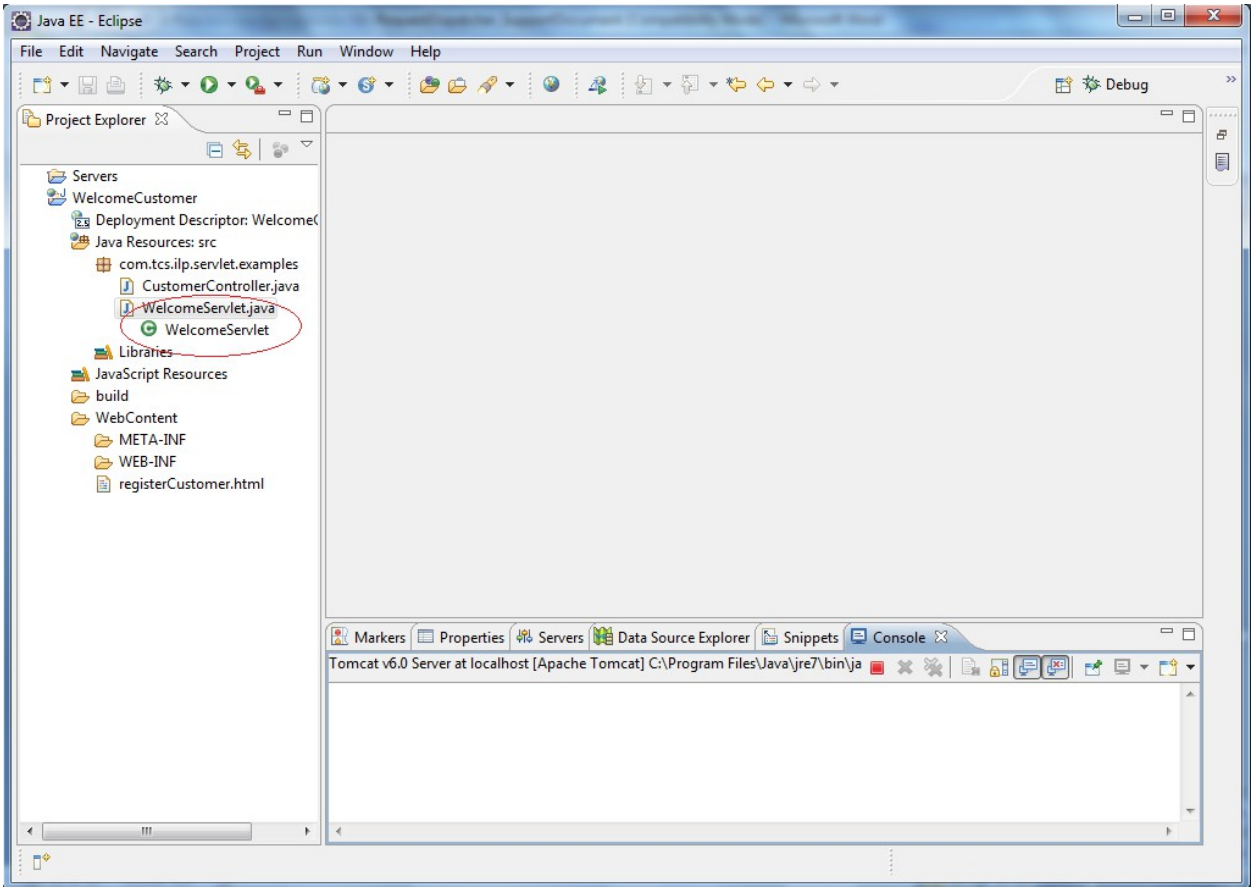
Create a servlet WelcomeServlet inside the same package as shown below



The “WelcomeServlet” holds the customized welcome message for the customer.

Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/WelcomeServlet.java - Eclipse

```
package com.tcs.ilp.servlet.examples;

import java.io.IOException;
import java.io.PrintWriter;

import javax.servlet.ServletException;
import javax.servlet.http.HttpServlet;
import javax.servlet.http.HttpServletRequest;
import javax.servlet.http.HttpServletResponse;

/**
 * Servlet implementation class WelcomeServlet
 */
public class WelcomeServlet extends HttpServlet {
    private static final long serialVersionUID = 1L;

    /**
     * @see HttpServlet#HttpServlet()
     */
    public WelcomeServlet() {
        super();
        // TODO Auto-generated constructor stub
    }

    /**
     * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse
     *      response)
     */
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        doPost(request, response);
```

Java EE - WelcomeCustomer/src/com/tcs/ilp/servlet/examples/WelcomeServlet.java - Eclipse

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Debug Java EE

WelcomeServlet.java

```java
    /**
     * @see HttpServlet#HttpServlet()
     */
    public WelcomeServlet() {
        super();
        // TODO Auto-generated constructor stub
    }

    /**
     * @see HttpServlet#doGet(HttpServletRequest request, HttpServletResponse
     *      response)
     */
    protected void doGet(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        doPost(request, response);
    }

    /**
     * @see HttpServlet#doPost(HttpServletRequest request, HttpServletResponse
     *      response)
     */
    protected void doPost(HttpServletRequest request,
            HttpServletResponse response) throws ServletException, IOException {
        // TODO Auto-generated method stub
        PrintWriter out = response.getWriter();
        String customerName = request.getParameter("Name");
        out.print("Welcome " + customerName);
    }

}
```

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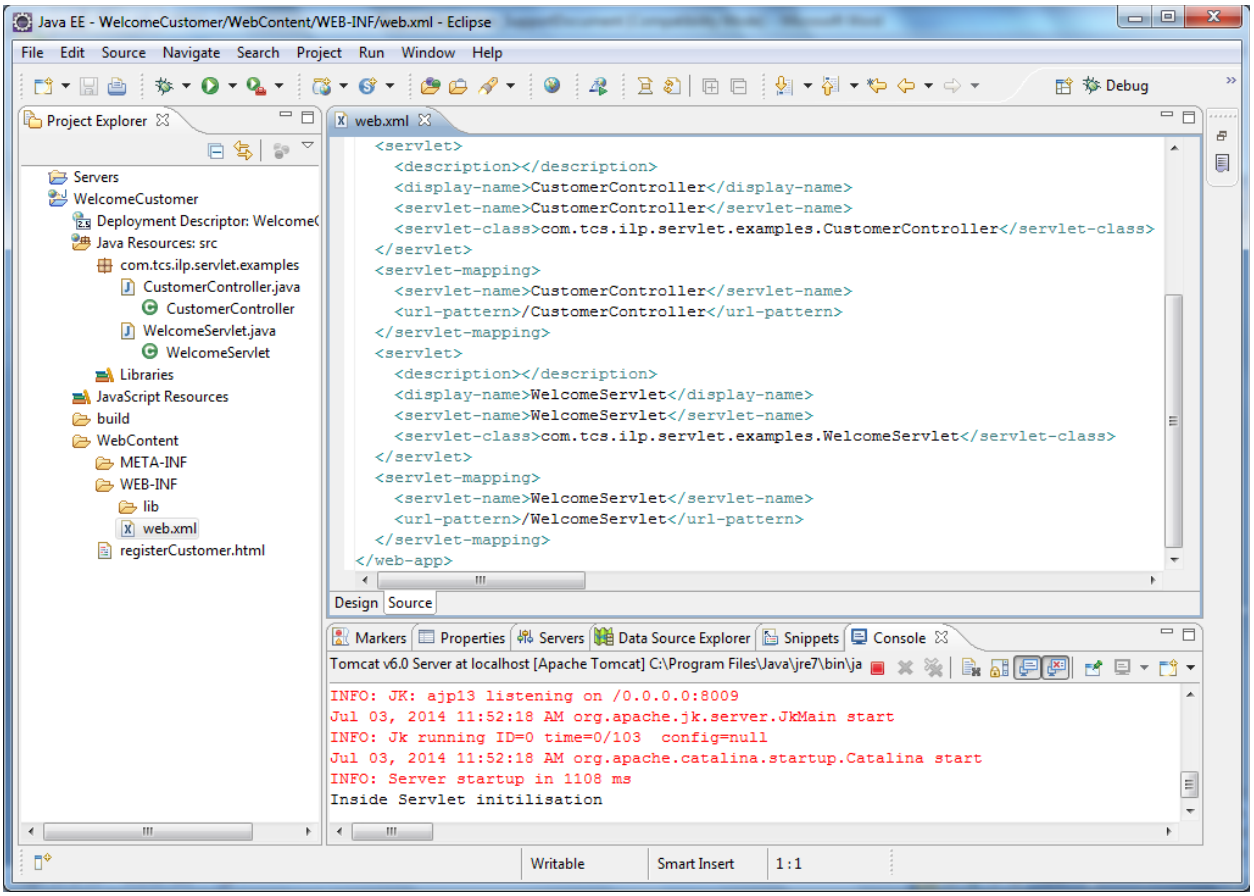
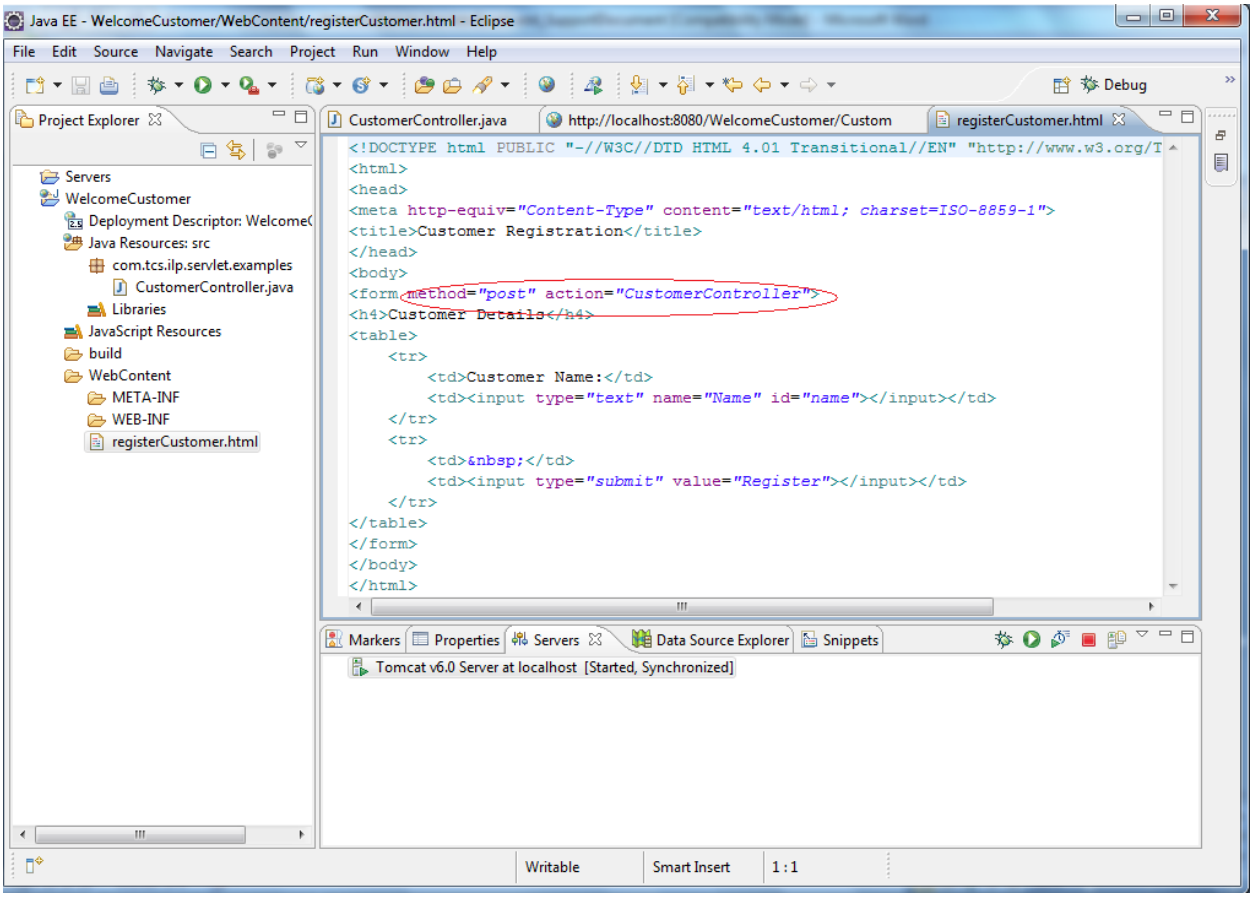
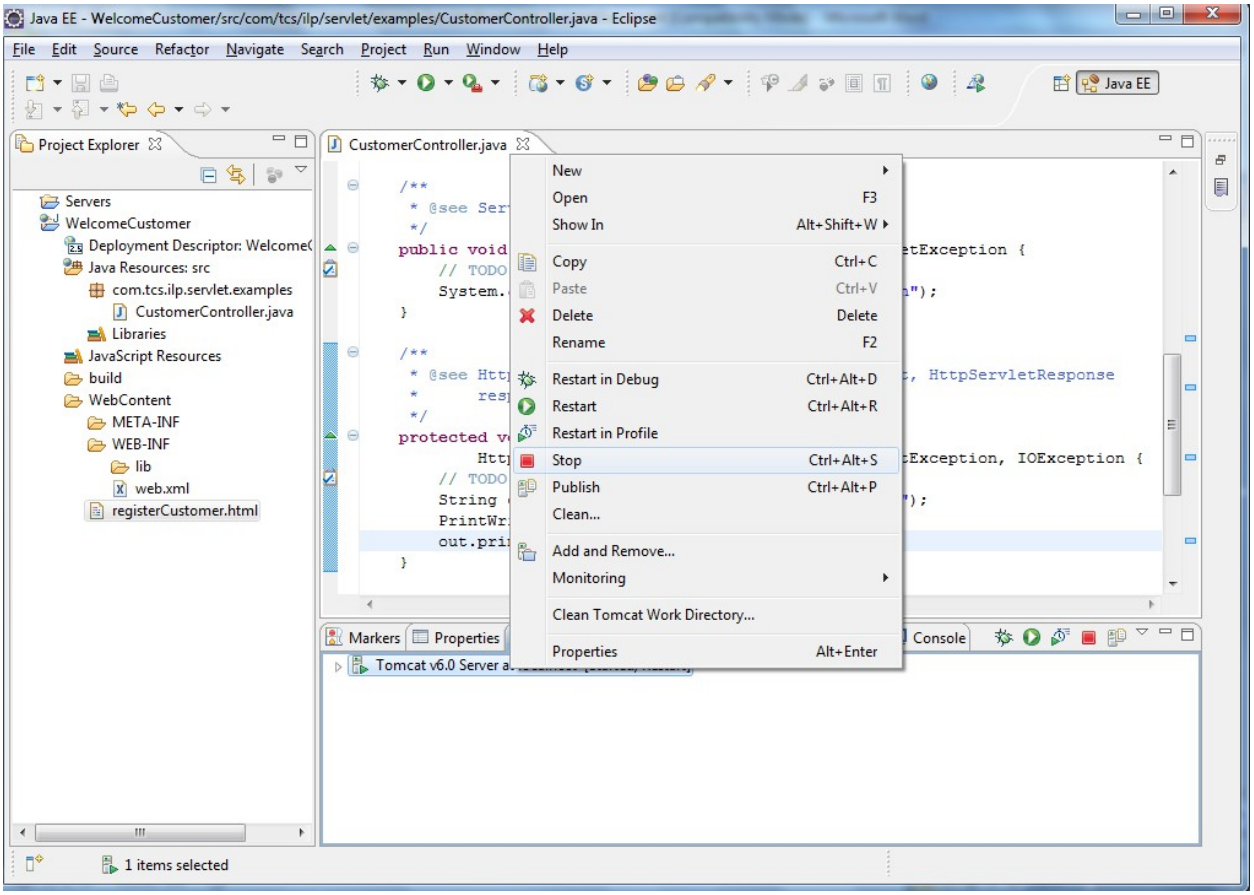
A corresponding mapping gets created in web.xml as the WelcomeServlet is created



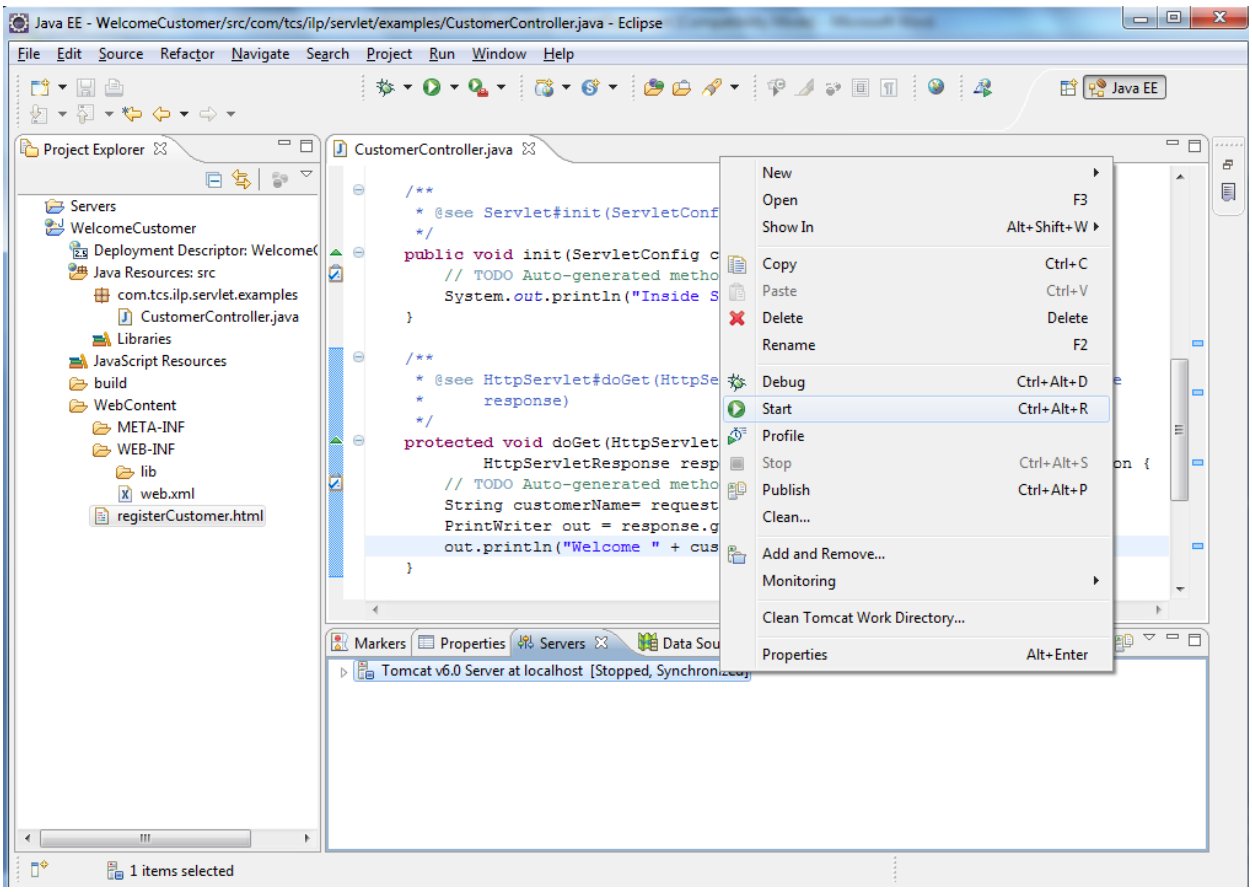
Provide the corresponding action and method for CustomerController inside registerCustomer.html



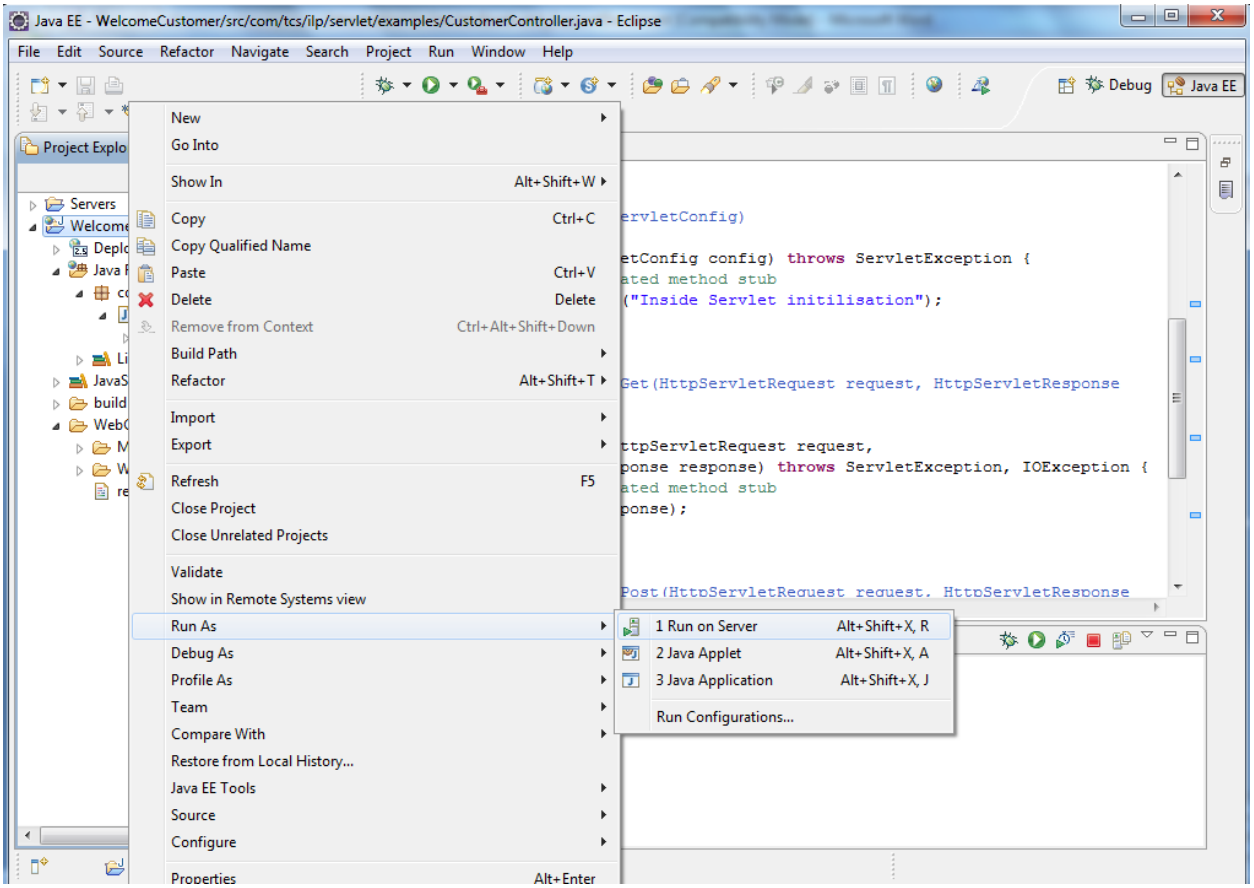
Create the server instance as in WebApplicationBasics_SupportDocument.doc.
Stop the server and restart the server again for the new changes to get reflected



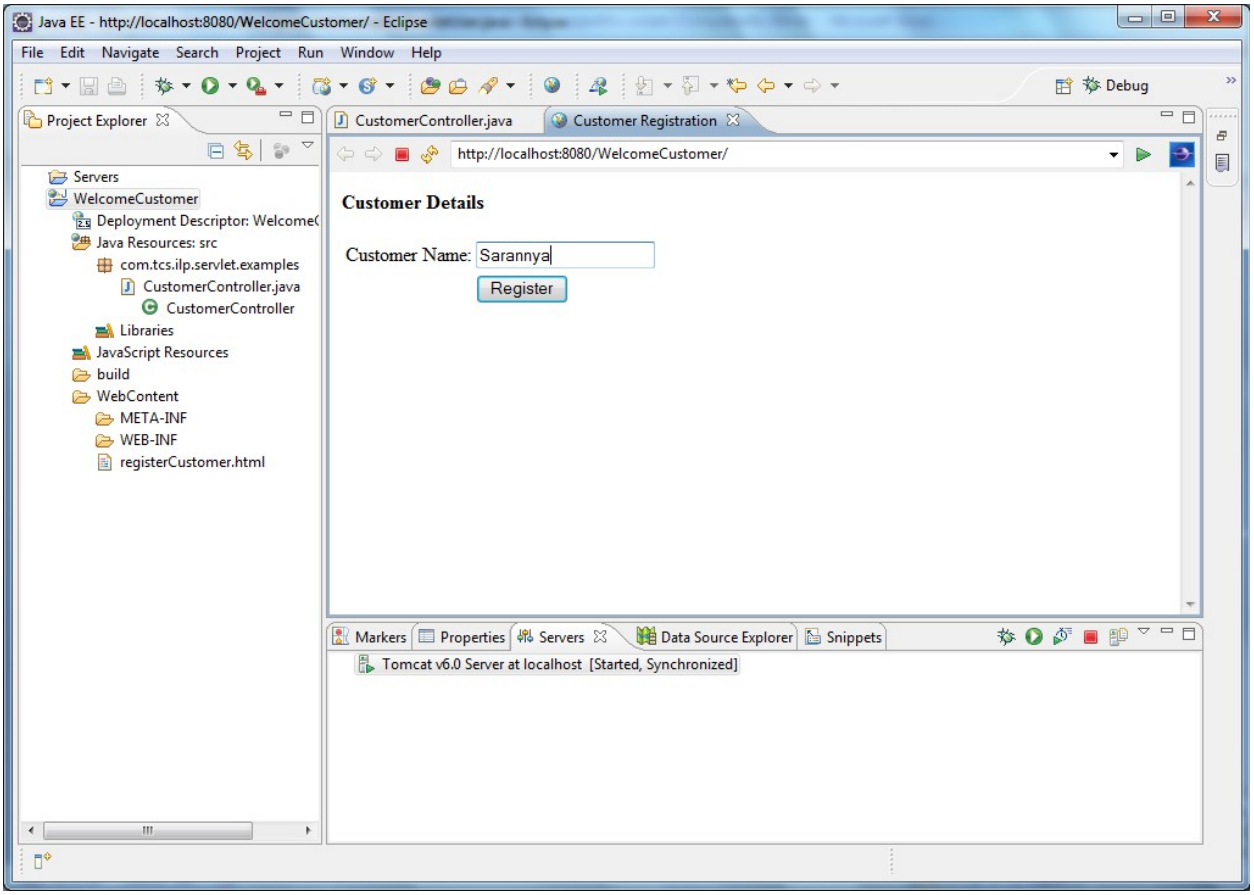
The server will be in stopped status. Now again restart the server.



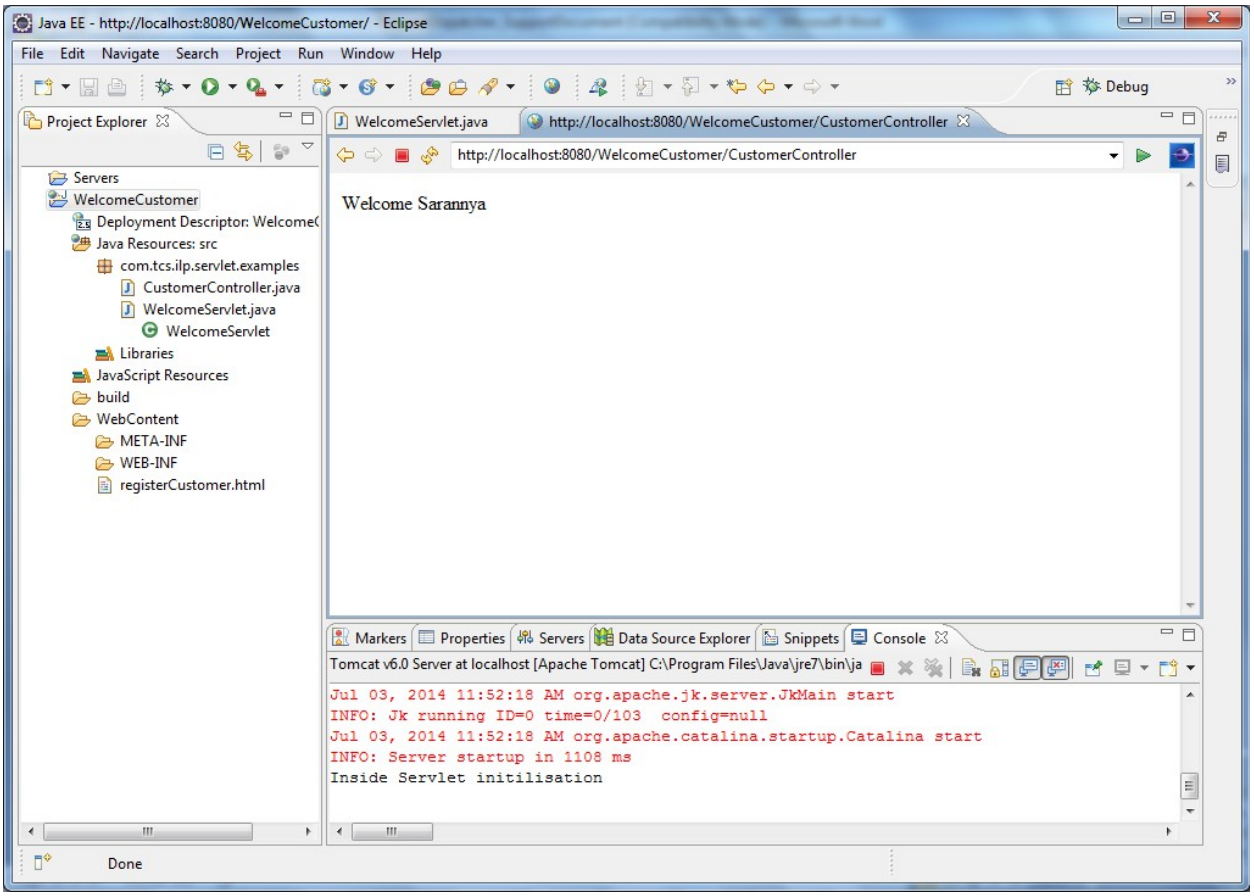
Once the server gets started, run the dynamic web project on server



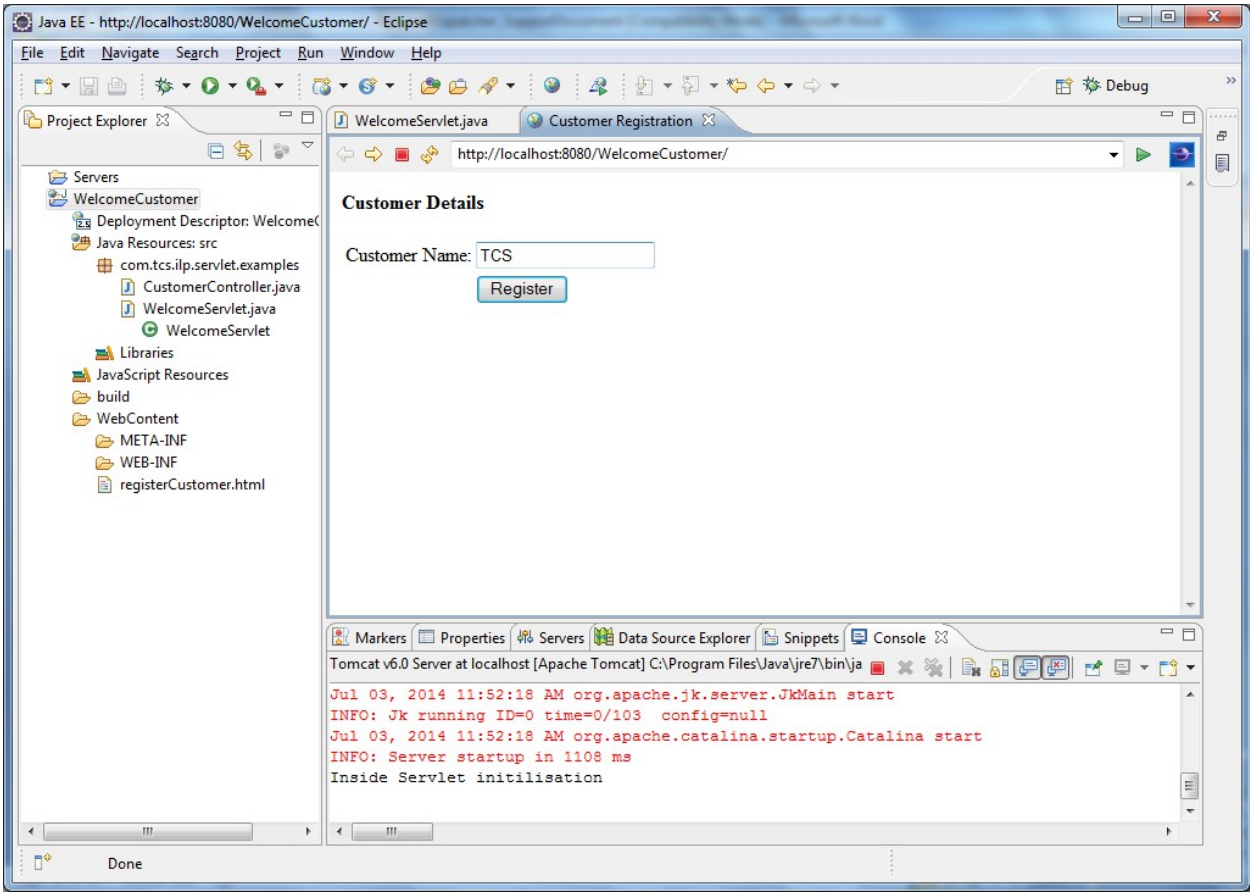
The output is shown on the screen. When a customer name is entered, the request gets Forwarded to the WelcomeServlet



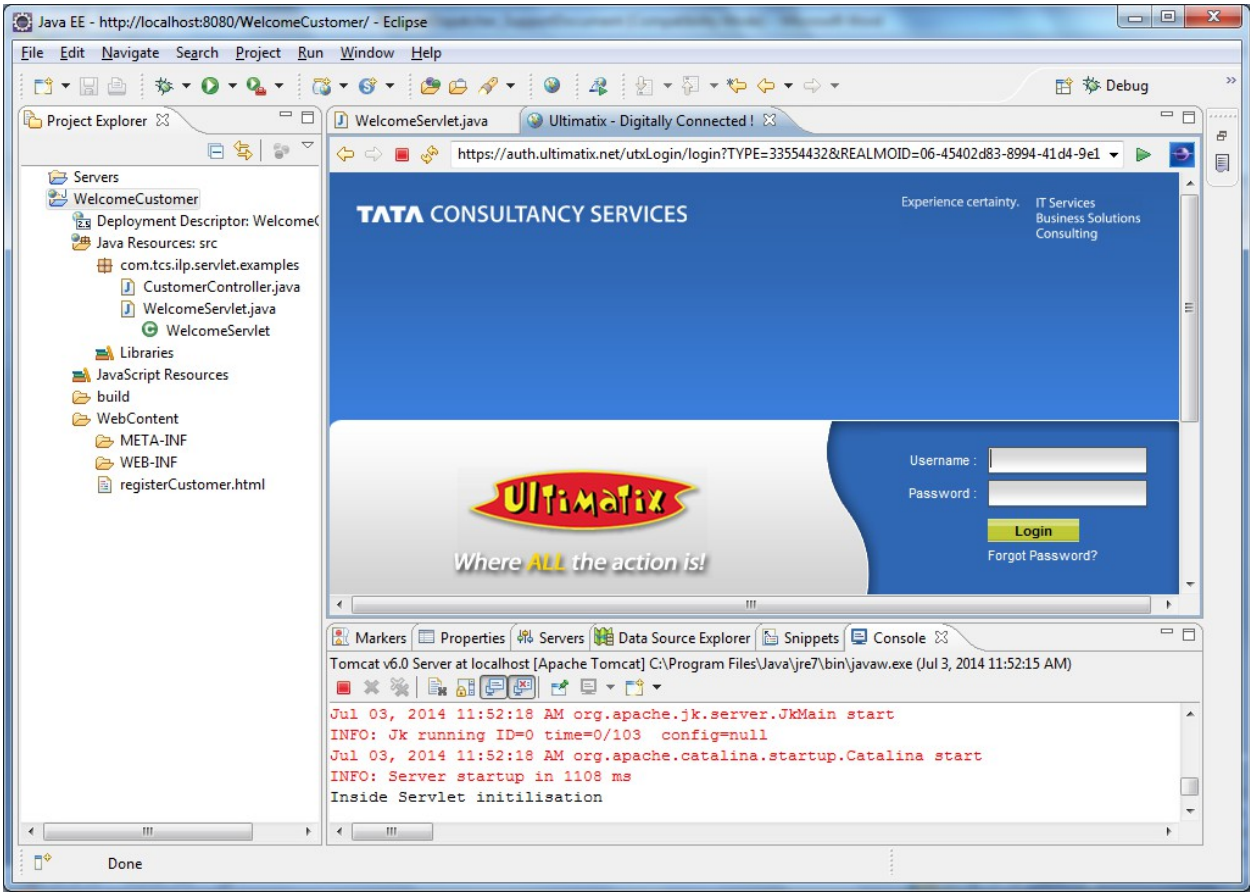
A welcome message with customer name gets displayed.



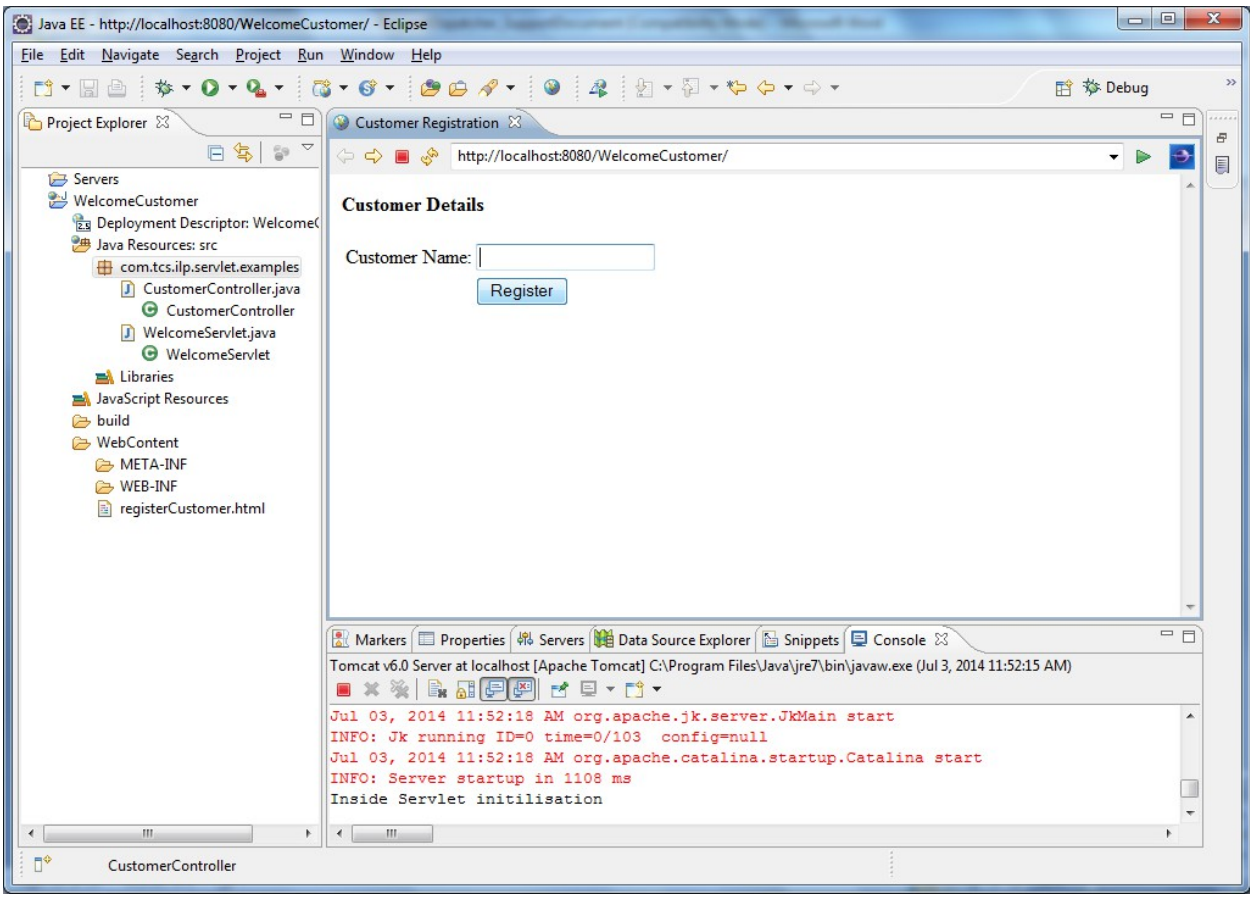
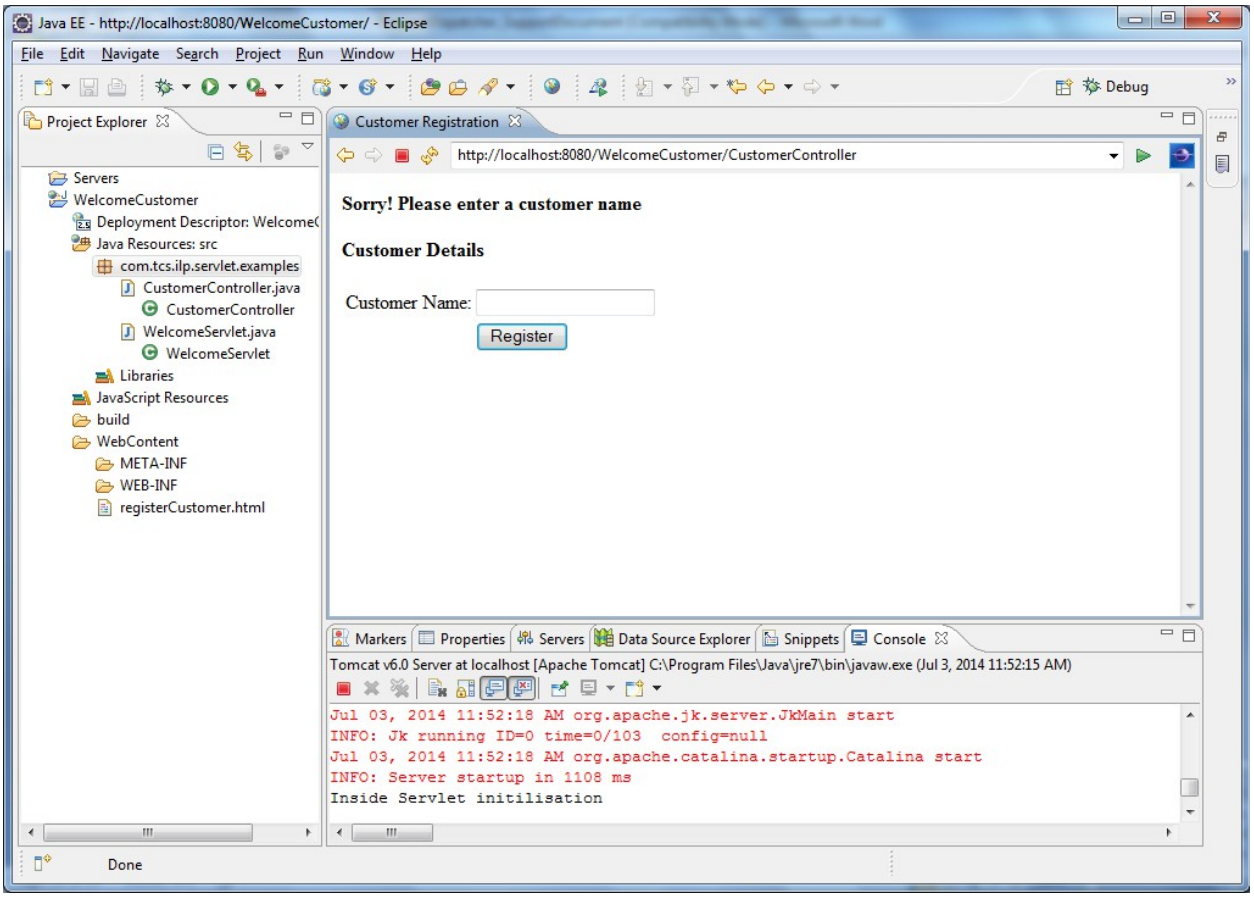
When the customer name is given as TCS, the request gets redirected to the ultimatix home page



Ultimatix home page getting displayed



When the customer name is empty, the request includes an error message and displays it on the same page



Error message getting included and displayed on the same page